

Scalability & Future Plan

Date	15 March 2026
Team ID	TM-RISINGWATERS-04
Project Name	Rising Waters: AI Powered Flood Prediction System
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	API Rate Limits on External Sentinel-2 Satellite Feeds	Restricts real-time flood mapping to static query intervals instead of an on-demand continuous sync.	High
2	In-Memory Thread-Based IoT Node Simulation	Limits testing to a small network of mock water-level nodes; cannot scale to hundreds of concurrent hardware feeds.	High
3	Single-Instance CPU Inference	Running deep learning UNet predictions on CPU limits processing speed, resulting in high latency during multiple basin evaluations.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Monolithic FastAPI server, capable of serving roughly 50 to 100 concurrent admin dashboard views.	Implement load balancing (NGINX) with auto-scaling Kubernetes pods and CDN caching for static assets.
2	Data Storage	Single local PostgreSQL/PostGIS database instance with basic spatial indices.	Transition to a distributed cloud database with read-replicas, partitioning tables chronologically, and hot-tiering metadata.
3	Performance	Single PyTorch thread processing predictions sequentially on CPU cores.	Deploy model inference pipelines on AWS SageMaker / GPU nodes utilizing Celery distributed task queues.
4	Security	Basic authentication (JWT) with direct publicly exposed backend endpoints.	Introduce an API Gateway layer with rate limiting, Web Application Firewall (WAF), and HTTPS end-to-end encryption.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Incorporate low-power LoRaWAN physical sensory nodes and optimize data telemetry.	Q3 2026	Enables off-grid, cost-effective field deployments across remote river basins.
Phase 3	Integrate Graph Neural Networks (GNNs) to model multi-basin topological water flow.	Q4 2026	Increases prediction accuracy of down-river flooding propagation to over 95%.
Phase 4		Q1 2027	

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
	Launch Native Android/iOS Alert apps with highly precise background geo-location tracking.		Minimizes communication overhead and delivers instant life-saving push notifications.